SAMITHA JASMINE

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PROFESSIONAL SUMMARY

Motivated Computer Science Engineering graduate seeking a **Python Developer (Fresher)** role, with a strong foundation in **Python, Object-Oriented Programming (OOP), Data Structures & Algorithms, REST APIs, SQL (MySQL), and Git**. Hands-on experience building, testing, debugging, and deploying **backend applications using Flask** with JWT authentication, CRUD operations, API integration, and relational databases, plus AI-powered applications using Gemini and Groq APIs. Strong analytical, problem-solving, and debugging skills.

SKILLS SUMMARY

Programming: Python, SQL, JavaScript

Backend & APIs: Python Backend Development, Flask, RESTful API Development, JWT Authentication, API Integration, CRUD Operations, Backend Development Concepts, Application Deployment (Gunicorn)

Frontend: HTML, CSS3, Bootstrap 5

Databases: MySQL, SQLite, ChromaDB

Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, SDLC, Requirements Analysis, Application Development, Debugging, Testing, Clean & Well-Documented Code, Coding Standards

AI & Automation: Generative AI, RAG Applications, Google Gemini API, Groq API

Tools & Practices: Git, GitHub, VS Code, Playwright (Fundamentals), Agile Development, Code Review

PROJECTS

AI Support Ticket Triage Agent

Python, Flask, SQLite, Google Gemini API, Bootstrap 5, Gunicorn

- Designed and developed an **AI-powered customer support backend application** using Python and Flask to automate ticket classification and workflow management.
- Analyzed application requirements and implemented automated classification of **ticket category, priority, and customer sentiment** using the Google Gemini API.
- Developed a **fallback rule-based mechanism** to improve reliability and maintain consistent classification when AI responses were unavailable.
- Built RESTful endpoints and an interactive dashboard with **dynamic filtering, workflow management, and automated analytics** to improve operational reporting.
- Tested and debugged application workflows to ensure reliable processing and consistent output.
- Integrated the application components, managed code with **Git/GitHub**, and deployed the solution using **Gunicorn**.

AI Document Chatbot

Python, Flask, REST API, SQLite, ChromaDB, Groq API, JWT

- Designed and built a **RAG-based AI application** using Python, Flask, REST APIs, ChromaDB, and Groq LLaMA 3.3 70B.
- Implemented **semantic document retrieval and context-aware response generation** to provide relevant answers across multiple documents.
- Developed secure **RESTful APIs with JWT authentication** and implemented per-user data isolation for secure application usage.
- Implemented custom hashing-based embeddings and semantic retrieval to optimize memory usage compared with standard vector storage approaches.
- Tested API workflows, authentication mechanisms, and data isolation to ensure reliable application behavior.
- Integrated multiple application components to deliver a complete AI-powered software solution.

Hospital Management System

Python, Flask, MySQL, HTML, CSS, JavaScript

- Designed and developed a **full-stack Hospital Management System** based on functional requirements for Admin, Doctor, and Patient workflows.
- Built and integrated **RESTful APIs and CRUD operations** using Python and Flask for efficient management of patient and appointment data.
- Implemented **role-based authentication and authorization** to secure application functionality and API endpoints.
- Designed **SQL** queries and relational database schemas in **MySQL** and integrated the Python backend with the web interface.
- Tested application workflows, debugged issues, and validated functionality across different user roles.
- Developed the solution from **requirement analysis through implementation and integration**.

INTERNSHIP

Full Stack Development Intern – The Mind IT

Trichy | July 2025

- Gained hands-on experience in **software development workflows, web application development, debugging, and industry development practices**.
- Worked under mentor guidance and followed structured development practices for implementing application functionality.
- Strengthened practical knowledge of frontend, backend, database, and application integration concepts.

EDUCATION

PGP College of Engineering and Technology | 2022 – 2026
Bachelor of Engineering in Computer Science and Engineering | CGPA: 7.98
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Web Technologies

CERTIFICATIONS

- HackerRank – Python (Basic), SQL (Basic), SQL (Intermediate)
- GUVI – AI Chatbot Development
- Udemy – The Ultimate Flask App
- Simplilearn – Basics of Web Scraping with Beautiful Soup for Beginners